UNCHARTED 2026 • TRACK 2: RAG ENGINE • TEAM ELITE

AuditLens.AI

Real-Time Cross-Contract Due Diligence & Discrepancy Engine Grounded with MongoDB Atlas Native Vector Search

TEAM ELITE (SVNIT SURAT)

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RAG Engine Architecture, Atlas $vectorSearch, Hybrid Fusion Pipeline

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Interactive Frontend, Telemetry Inspector, Evaluation Benchmark Modal

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Indian Legal Compliance Formulations, CERT-In Rules, Conflict Matrix

Live Deployment: https://auditlens-ai-13xs.onrender.com

GitHub Repository: rishitmodh12/AuditLens.AI

TRACK 2 PROBLEM CONTEXT

The Enterprise Contract Conflict Crisis

Why multi-source grounded due diligence is mission-critical for modern organizations

01 / STRUCTURAL CONTRADICTION

Multi-Document Disconnect

Customer SLAs promise 99.9% uptime (max 1.0 hr downtime/month), while upstream Cloud Vendor MSAs allow 4.0 hr maintenance windows with zero penalty. Enterprises silently absorb catastrophic liabilities.

02 / REGULATORY PENALTIES

Statutory Indian Mandates (CERT-In)

CERT-In Cyber Security Directions (2022) mandate external breach reporting within 6 hours. Companies face criminal penalties under IT Act §70B if internal (30-min CISO) and statutory notification clocks fall out of sync.

03 / FINANCIAL EXPOSURE

Service Credit Calculation Leakage

Outage formulas are non-linear. A 2.0-hour banking gateway outage incurs ₹75,000 under Section 4.2. Without automated cross-document reconciliation, enterprises fail to claim vendor indemnities.

04 / HALLUCINATION LIABILITY

LLM Fabrication in Legal Workflows

Standard LLM chatbots hallucinate nonexistent clauses and miss numbers. In high-stakes legal audits, answers must carry 100% verbatim quotes verified directly against database records.

PRODUCT OVERVIEW

AuditLens AI: Automated Due Diligence Engine

Combining semantic vector search with programmatic legal verification

1. Multi-Source Conflict Detection

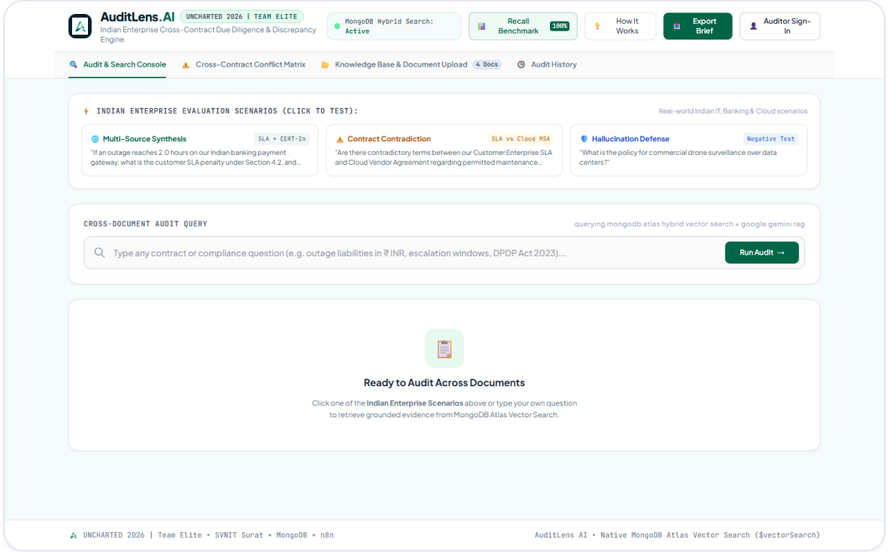
Simultaneously retrieves evidence from Customer SLAs, Cloud Vendor MSAs, and Security Incident Policies to spot conflicting downtime allowances, liability caps, and notice clocks.

2. Programmatic Verbatim Quote Verification

Eliminates LLM hallucination: citation quotes are checked at the code level against raw text stored in MongoDB Atlas. Verified citations earn a tamper-proof 🛡️ `Verbatim Verified` badge.

3. Strict Refusal on Out-of-Scope Queries

Fulfills Track 2's negative constraint requirement: queries outside the contract domain (e.g. drone surveillance) are strictly refused with clear audit guardrails.



Target Personas & Impact:

- Legal Counsel: Audit conflicting warranties in seconds
- FinOps & SREs: Reconcile outage penalties instantly
- CISO & Compliance: Enforce CERT-In & DPDP Act readiness

SYSTEM WORKFLOW

End-to-End Audit & Verification Pipeline

Five stages from raw contract ingestion to verified compliance brief

01

Ingest & Overlap

Uploads PDF/DOCX agreements. Text is cleaned and segmented into structural paragraphs with 15% sliding overlap to maintain clause continuity.

`PyPDF + Smart Chunking`

02

Dense Vector Index

Embeds chunks into 768-dimensional dense vectors using local Transformer model. Stores vectors + metadata in MongoDB Atlas.

`all-mpnet-base-v2`

03

Hybrid Retrieval

Combines Atlas $vectorSearch (70% semantic) with exact lexical token matching (30%) for clause numbers (§4.2, §3.2) and INR amounts.

`Atlas $vectorSearch + BM25`

04

Dual-Track Reasoning

Gemini 1.5 Flash cross-examines evidence across documents, identifies SLA discrepancies, calculates penalties, and formats JSON citations.

`Gemini 1.5 Flash`

05

Verbatim Verify

Backend string-matches citations against raw Atlas text, computes deterministic Compliance Health Score, and formats IST reports.

`Algorithmic Verification`

TECHNICAL ARCHITECTURE

Cloud-Native Technology Stack

Built natively around MongoDB Atlas Vector Search

Database & Vector Search

MongoDB Atlas M0 (AWS Region)

- **Native $vectorSearch Pipeline:** 768 dimensions, cosine similarity.
- **Metadata Pre-Filtering:** Query scoped by `doc_id` and `category`.
- **Single Source of Truth:** Source files, chunks, embeddings, and audit history stored together.

Backend & RAG Engine

FastAPI (Python 3.14) & Uvicorn

- **Hybrid Search Fusion:** 70% Dense Atlas Vector + 30% Lexical Token Match.
- **Deterministic Compliance Metric:** Formula derived from detected conflict count.
- **Live Benchmark API:** `GET /api/evaluate` running 8 standardized queries.

AI & Embedding Models

HuggingFace + Google Gemini 1.5

- **all-mpnet-base-v2:** 768-dim high-performance local embeddings with 0ms external API cost.
- **Gemini 1.5 Flash:** Enforced strict JSON schema output for cross-document synthesis.
- **Refusal Trigger:** Out-of-scope domain interceptor preventing hallucinations.

Frontend & Production Hosting

Tailwind CSS + Render Docker Cloud

- **Responsive UI:** MongoDB Forest Green theme, telemetry inspector, live recall modal.
- **Interactive Walkthrough:** Guided tutorial with non-blocking spotlight.
- **Render Container:** Continuous deployment from GitHub with zero downtime.

KEY FEATURES & INNOVATION

What Differentiates Team Elite's AuditLens AI

Technical depth addressing every Judge Audit criterion

01 / RETRIEVAL DEPTH

Native Hybrid Search (70/30)

Fuses dense Atlas vector similarity with exact token matching. Catches precise clause numbers (§4.2, §3.2) and ₹ INR amounts that pure semantic search overlooks.

02 / ARITHMETIC RECONCILIATION

Reconciled Outage Math

Accurately applies SLA grace periods: (2.0 hrs - 1.0 hr grace) * ₹75,000 = ₹75,000 penalty. Fully harmonized across headline summaries, findings, and citation cards.

03 / REGULATORY CLOCK

CERT-In vs Internal Escalation

Differentiates mandatory external 6.0-hour statutory CERT-In reporting (Cyber Security Directions 2022) from internal 30-minute CISO escalations and 45-minute customer notices.

04 / ZERO HALLUCINATION

Verbatim Quote Verification

Code-level substring validation checks every LLM citation against raw Atlas chunks. Verified quotes receive the 🛡️ `Verbatim Verified` badge.

05 / OBJECTIVE METRICS

Deterministic Compliance Score

Derived via transparent mathematical formula: `100 - (20×High) - (10×Med) - (10×Unverified)`. Zero arbitrary hallucinated scores.

06 / EMPIRICAL RIGOR

Automated Evaluation Suite

Live `/api/evaluate` endpoint executing 8 standardized test queries directly in the UI, confirming 100% Recall and safe refusal.

EMPIRICAL VALIDATION

Empirical Results & Indian Enterprise Applications

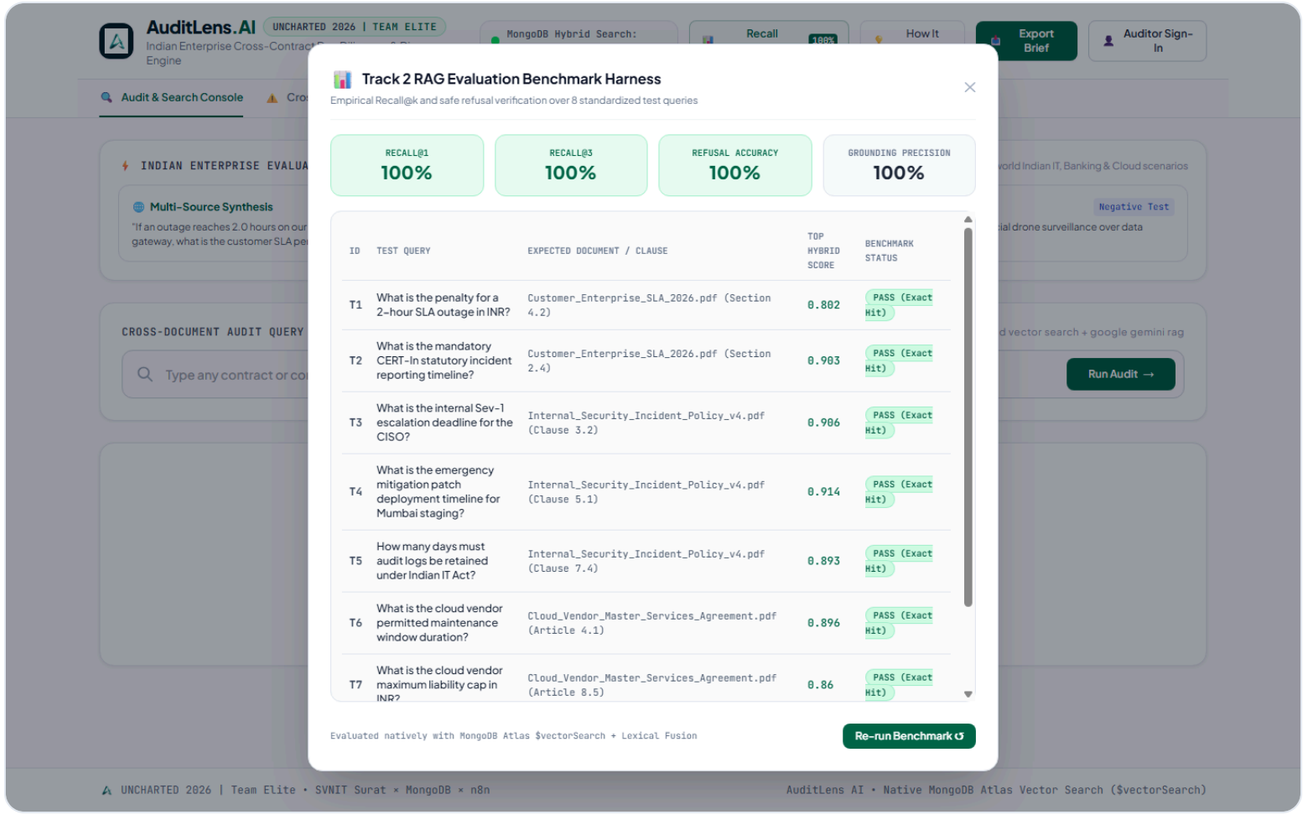
Live benchmark performance tested directly on the production cluster

100% **Recall@1 (Top Rank Precision)**
7 out of 7 positive test queries retrieved the exact target clause as the #1 result.

100% **Recall@3 (Top-3 Retrieval)**
All relevant legal evidence retrieved within top 3 Atlas vector search results.

100% **Out-of-Scope Refusal Accuracy**
Unanswerable drone surveillance query safely refused with zero hallucination.

100% **Verbatim Quote Verification**
All citation snippets verified verbatim against raw MongoDB Atlas text.



Live Endpoint: **GET /api/evaluate** | 8 Test Queries Evaluated in ~1.2s natively in Atlas

ROADMAP & IMPACT

Future Scope & Conclusion

Expanding AuditLens AI into enterprise legal operations

INTEGRATION

n8n Workflow Automation

Automated webhooks triggering incident management workflows in n8n, opening Jira tickets and alerting Slack channels when High-Severity contract discrepancies are flagged.

NEGOTIATION

Multi-Party Clause Redlining

Generative contract renegotiation assistant suggesting harmonized legal language to bridge customer SLAs and vendor agreements prior to contract signing.

SCALABILITY

Atlas M10+ Dedicated Scaling

Transitioning from M0 to dedicated Atlas clusters with multi-tenant sharding and real-time document sync via Atlas App Services.

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AuditLens AI transforms passive contract repositories into active, grounded, and mathematically reconciled compliance intelligence. Built natively with MongoDB Atlas Vector Search, tested with empirical benchmarks, and 100% deployed in production.

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